

Recursive Fractal Cosmology:

A Generative Ontology of Existence

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May 2026

Boundary: RFC is internally coherent as a deterministic triadic derivation system with a growing first-pass validation scaffold. It is not yet fully externally validated as a complete laboratory theory of all physical constants and observables.

Abstract

Recursive Fractal Cosmology (RFC) is developed here as a triadic account of existence in which physical reality emerges through the interaction of the Cosmic Infinite Field (CIF), the Quantum Vacuum (QV), and the Recursive Fractal Lattice (RFL). Rather than treating cosmology as a fitted collection of independent modules, RFC is framed as a generative ontology: CIF supplies modal possibility, QV supplies recursive compression and selection, and RFL supplies stabilized structure, memory, geometry, and observability.

This paper presents the deterministic G/R/N/S/T reconstruction and extends it with a staged downstream validation layer. Module U gives a first-pass one-anchor electromagnetic and atomic constant table. Module V gives a compressed cosmology screen. Module W gives a light-abundance BBN proxy. Module X gives CP/EDM bound and baryon-eta proxies. Module Y2 gives an exploratory particle-sector refinement map. Module Z gives observer, branching, neural, and EEG target screens. Module QG gives a finite spin-foam transition-amplitude audit.

The paper is careful about claim boundaries. Module U is treated as a completed first-pass screen, while Modules V, W, X, Z, and QG are partial proxy or finite-audit screens. Module Y2 is explicitly exploratory because it uses reference values for map selection; it must be frozen and retested as a future Y3 module before independent validation claims are made. RFC is therefore presented not as a final replacement for established physics, but as a reproducible triadic derivation program with a growing validation scaffold.

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1. Introduction

RFC begins from the claim that existence is structured before it is conventionally physical. A particle already assumes localization, a field already assumes a domain of variation, and spacetime already assumes ordered relation. RFC therefore asks what generative process could make stable physical structure possible at all.

The answer proposed here is triadic recursion. CIF, QV, and RFL form the minimal grammar of emergence. CIF is unbounded modal possibility. QV is recursive compression, damping, entropy flow, and selection. RFL is stabilized manifestation, geometry, memory, observability, and physical structure. In this formulation, physical law is not imposed from outside the system; it is the stable behavior of recursive emergence itself.

The current framework removes old upstream fitting from the derivation chain. External references appear at the end of the chain, as comparison and validation, not as upstream parameter selectors.

Note: Current evaluation order: ontological coherence -> mathematical derivation -> deterministic closure -> dimensional projection -> dimensionless validation -> SI bridging -> downstream validation screens -> full external empirical validation.

2. Ontological Foundation: CIF, QV, and RFL

The CIF-QV-RFL triad is not a decorative metaphor in the current framework. It is the root dependency structure of RFC. Every later module is interpreted as a downstream expression of this triadic grammar.

Triad element	Meaning	Role in RFC
CIF	Cosmic Infinite Field	Unbounded modal possibility; the reservoir of possible structure before selection.
QV	Quantum Vacuum	Recursive compression, damping, entropy flow, asymmetry, and branch selection.

Triad element	Meaning	Role in RFC
RFL	Recursive Fractal Lattice	Stabilized recursive manifestation: geometry, memory, observability, and physical structure.

$$QV(CIF) \longrightarrow RFL$$

The theory is triad-first: later cosmological, particle, quantum-geometric, observer, neural, and collapse-rebirth modules are downstream projections, not independent foundations.

3. Big Implosion and the Collapse-Rebirth Cycle

The Big Implosion is the first movement of the triad. Cosmogenesis does not begin as a simple outward explosion from a pre-existing spacetime. It begins as inward recursive compression: QV acts on CIF and produces the first RFL structure.

Cycle phase	Role in RFC
Implosion	QV compresses CIF potential into first RFL structure.
Emergence	Symbolic and physical field structures activate.
Expansion	Instantiated RFL develops large-scale structure and domain projections.
Saturation	Recursive compression approaches a limit and the system accumulates memory.
Meltdown	Recursive coherence begins to fail and established attractors destabilize.
Reset	CIF remains as modal reservoir and the cycle can reinitialize.

In this interpretation, dark matter is an early compression relic, while dark energy is a late dissipative tail. They are two temporal expressions of the same recursive process: early QV compression and late kernel-tail persistence.

4. Recursive Entropy, Time, and Observer Identity

RFC treats recursive entropy as memory-gradient rather than mere disorder. Entropy tracks how CIF possibility becomes differentiated, compressed, and stabilized through QV into RFL structure.

$$S_{\text{rec}}(t) = \sum_{j=1}^n j \ln(\delta) \delta^{-j} e^{-\alpha j t}$$

Time is associated with this entropy gradient. It is not only a container in which recursion occurs; it is the measurable slope of recursion itself. Observer identity emerges when an RFL structure can preserve phase coherence through entropy flow, contradiction, bifurcation, and branch divergence.

$$\lambda_{\text{div}}(t) = \ln(1 + |\psi_1(t) - \psi_2(t)|)$$

5. Recursive Kernel and Symbolic Field Architecture

The shared recursive kernel is the formal expression of triadic action. It represents QV compression acting on CIF modal content to produce RFL attractors.

$$K_f(t) = \sum_{j=1}^n \delta^{-j} f_j(t) e^{-\alpha j t}$$

$$\psi(t; \nu) = \sum_{j=1}^n \delta^{-j} \cos(jt + \nu) e^{-\alpha j t}$$

$$E(t) = \left(\frac{d\psi}{dt}\right)^2 + \psi(t)^2 \quad m(\nu) = \sqrt{\langle \psi(t; \nu)^2 \rangle}$$

Different domain modules instantiate different modal bases, but the operator remains shared. The same kernel underlies cosmology, CP/asymmetry, symbolic mass generation, observer divergence, collapse-rebirth memory, and quantum-geometry proxies.

6. Replacement of Old Module G

The central methodological change is the removal of old MCMC/NUTS parameter assimilation from active RFC derivation. That earlier approach was useful during exploration, but it is incompatible with the mature claim that foundational parameters should be generated internally by the triad.

Note: Old developmental flow: observational targets -> MCMC/NUTS -> fitted parameters -> modules. Current active flow: CIF-QV-RFL triad -> recursive kernel -> deterministic closure -> frozen packet -> modules -> validation.

The older version remains historical development context. The active Module G is deterministic triadic closure. The frozen packet is generated before downstream comparison and is not retuned by the validation screens.

7. Module G: Deterministic Triadic Closure

The active Module G derives the internal packet from closure requirements of the CIF-QV-RFL triad and uses no empirical targets. This packet is frozen for all downstream screens in this paper.

$$\alpha = \frac{\ln(\delta)}{60} \qquad \nu = k\delta^{-4} \qquad \varepsilon = \alpha \nu$$

Quantity	Value	Role
δ	4.6692	Recursive bifurcation scale.
cycle length	60	Canonical recursive cycle length.
α	0.0256831	QV damping scale; $\ln(\delta)/60$.
phase depth k	2	Stable CIF phase-depth branch.
ν	0.00420784	CIF phase seed; $k\delta^{-4}$.
ε	0.000108071	QV-CIF asymmetry scale; $\alpha\nu$.
λ normalized	0.489442	RFL nonlinear stability coupling.
n closure	18	Effective recursive closure depth.
n full canonical	40	High-resolution modal basis.
empirical targets used	False	Confirms non-MCMC derivation.

Deterministic check	Absolute error
$\alpha = \ln(\delta) / \text{cycle length}$	2.91799e-8
$\nu = k\delta^{-4}$	8.43445e-9
$\varepsilon = \alpha\nu$	6.24496e-10

Closure channel at n = 18	Value
QV	0.144904
CIF	0.00100137

Closure channel at n = 18	Value
RFL normalization	0.727294
CP	0.00856146
Collapse-rebirth	0.280559
Recursive tail	5.9996×10^{-26}

8. Module R: Triad-Grouped Global Closure Audit

Module R tests whether the frozen Module G packet remains globally coherent across downstream source structure. It does not retune the packet. It audits whether the apparent RFL residual is source-coupled to QV, CIF, RFL, CP, dark-kernel, observer, and collapse-rebirth channels.

$$R_{\text{raw}} = \left\langle \frac{(\ddot{u} + \lambda u^3 - u)^2}{1 + u^2} \right\rangle$$

$$R_{\text{source}} = R_{\text{raw}} - \sum_i w_i S_i(t)$$

Audit value	Result
raw standardized residual score	0.990099
source-coupled RFL residual score V2	0.576064
residual improvement V2	0.418175
Module R score V2	0.726401
CP residual	3.59005 x 10^-10
best lag	-4
best lag correlation	0.873361
memory transfer score	0.146153
tail N18	5.9996 x 10^-26
tail N40	2.1451 x 10^-55

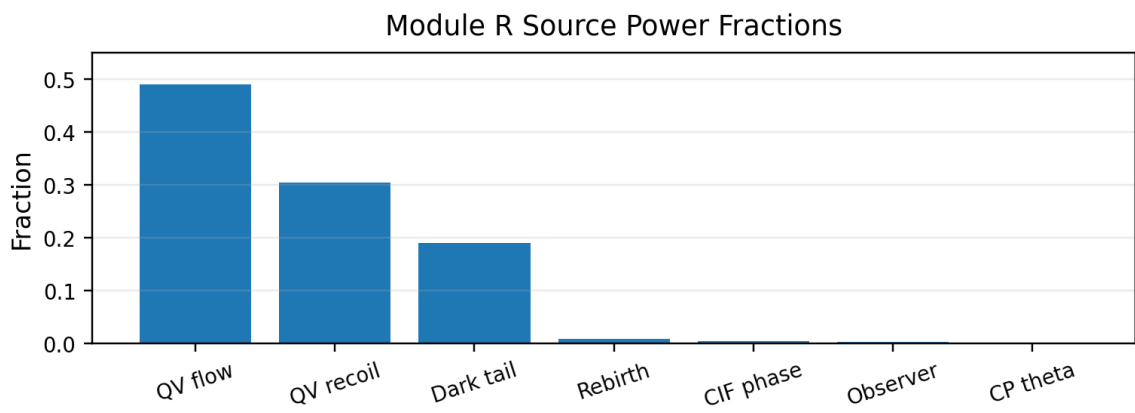


Figure 1. Module R source power fractions. QV entropy flow, QV recoil, and dark-kernel tail dominate the closure audit.

The key result is not a single low residual. The key result is the structure of the residual: it improves substantially when treated as source-coupled, and its grouped power is overwhelmingly QV-dominant.

9. Module N V2: Dimensional Projection Bridge

Module N V2 corrects the earlier projection problem by making projection dimension-aware. It separates symbolic constants, internal projected units, one-anchor SI bridging, and dimensionless coupling mapping.

Projection bridge quantity	Value
energy factor	0.28443
time factor	1.54099
cycle time projected	92.4593
symbolic mass projected	0.0465119
mean energy projected	0.00438067
α_{EM} projected inverse-energy	228.276
α_G projected mass-squared	0.00216335
Λ projected inverse-energy-cycle	2.46893

$$\alpha_G \approx m^2 \qquad \alpha_{EM}E \approx 1 \qquad \Lambda ET \approx 1$$

Identity residual	Value
mass-squared identity	0 to 4.33681×10^{-19}
inverse-energy identity	2.28276×10^{-10}
lambda-cycle identity	2.4688×10^{-12}

Note: The raw projected inverse-energy value near 228 is not claimed to be the physical inverse fine-structure constant. Module T performs the separate dimensionless coupling map.

10. Downstream Physical-Projection Screen

The downstream physical-projection screen tests whether the frozen G/R/N packet can support RFC domain projections without retuning. It covers dark-sector behavior, CP/asymmetry, primordial-abundance proxy behavior, observer branching, collapse-rebirth memory transfer, and geometry coherence.

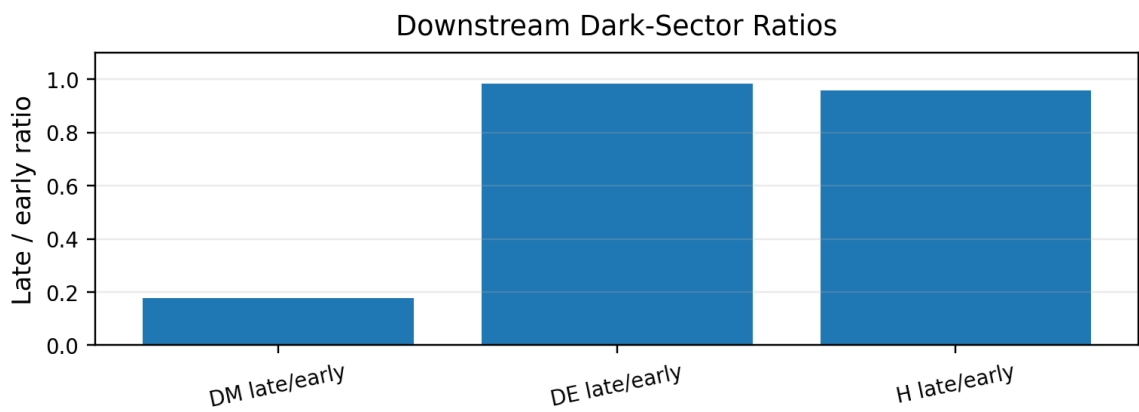


Figure 2. Dark-matter compression decays while the dark-energy tail persists.

Domain	Quantity	Value	Interpretation
Dark matter	early compression	0.216534	Early compression relic.
Dark matter	late compression	0.0381948	Late compression is much lower.
Dark matter	decay ratio	0.176392	Compression decays strongly.
Dark energy	early tail	2.52655	Persistent tail term.
Dark energy	late tail	2.48382	Tail remains high.
Dark energy	tail ratio	0.983088	Tail persists across cycle.
Hubble proxy	late/early ratio	0.958858	Mild decay; no blow-up.
BBN proxy	final Yp	0.249278	Light-abundance proxy only.
CP/EDM	CP residual	3.59079 x 10 ⁻¹⁰	Stable asymmetry residual.
Observer	decoherence proxy	5.73826 x 10 ⁻⁸	Controlled branch divergence.
Rebirth	best lag correlation	0.905876	Strong lagged memory transfer.

Domain	Quantity	Value	Interpretation
Geometry	mean coherence	0.940535	High geometry coherence.

11. Dimensionless Observable Validation Layer

The dimensionless validation layer tests ratios, residuals, identity checks, coherence measures, suppression factors, and stability flags before SI calibration and before external data comparison.

Validation flag	Status	Criterion
Module N identities preserved	True	$\alpha_G/m^2 = 1$; $\alpha_{EM} E = 1$; $\Lambda ET = 1$.
Dark matter decays	True	DM late/early = 0.176392.
Dark energy tail persists	True	DE late/early = 0.983049.
CP residual tiny	True	$R_{CP} = 3.59005 \times 10^{-10} < 10^{-8}$.
Collapse-rebirth memory strong	True	Best lag correlation = 0.873361 > 0.8.
Observer branching controlled	True	Observer divergence = $7.48103 \times 10^{-6} < 10^{-4}$.
Geometry coherence high	True	Mean geometry coherence = 0.94051 > 0.9.
overall validation score	1	All seven flags pass.

12. Module S: One-Anchor SI / Laboratory Bridge

Module S introduces a one-anchor SI bridge. The electron rest energy is used only as a unit conversion anchor. It is not treated as a prediction and does not retune Modules G, R, or N.

Module S field	Value
one anchor used	True
anchor choice	electron rest energy
parameter search performed	False
Module G/R/N retuned	False
SI bridge quantity	Value
internal RFC mass-like unit	0.0463552
1 RFC energy unit	11.0236 MeV
1 RFC energy unit	$1.76617 \times 10^{-12} \text{ J}$
1 RFC time unit	$5.97096 \times 10^{-23} \text{ s}$
1 RFC length unit	$1.79005 \times 10^{-14} \text{ m}$
projected cycle time	$5.52071 \times 10^{-21} \text{ s}$
projected cycle length	$1.65507 \times 10^{-12} \text{ m}$
anchor-check mass-energy	0.510999 MeV
anchor-check mass	$9.10938 \times 10^{-31} \text{ kg}$

The anchor check recovers the electron mass-energy because electron rest energy was chosen as the single unit anchor. This is a calibration check, not an independent prediction.

13. Module T: Dimensionless Coupling Map

Module T maps the raw Module N inverse-energy quantity into a dimensionless fine-structure-like coupling using a triad-derived screen factor.

$$\alpha_T^{-1} = \alpha_N^{-1} Q_V \sqrt{\lambda_{\text{norm}} Q_{\text{memory}}} [1 - (P_{RFL} - P_{CIF})]$$

Module T input	Value
raw inverse-energy coupling	228.807
QV inclusive	0.984868
λ normalized	0.489442
Q memory	0.761993
RFL power	0.010047
CIF power	0.00508468
screen factor	0.598472
mapped inverse coupling	136.935
reference comparison	137.036
relative difference	0.000740478 = 0.0740478%

$$228.807 \times 0.598472 = 136.935$$

Note: Correct interpretation: 228.807 is an internal inverse-energy value that becomes a fine-structure-like dimensionless value only after the Module T map. Incorrect interpretation: 228.807 equals the physical inverse fine-structure constant.

14. Extended Downstream Validation Screens

The downstream validation layer follows Module T. These screens do not erase the remaining need for peer review and domain-complete external validation. They do, however, move RFC beyond internal reconstruction into a staged validation program.

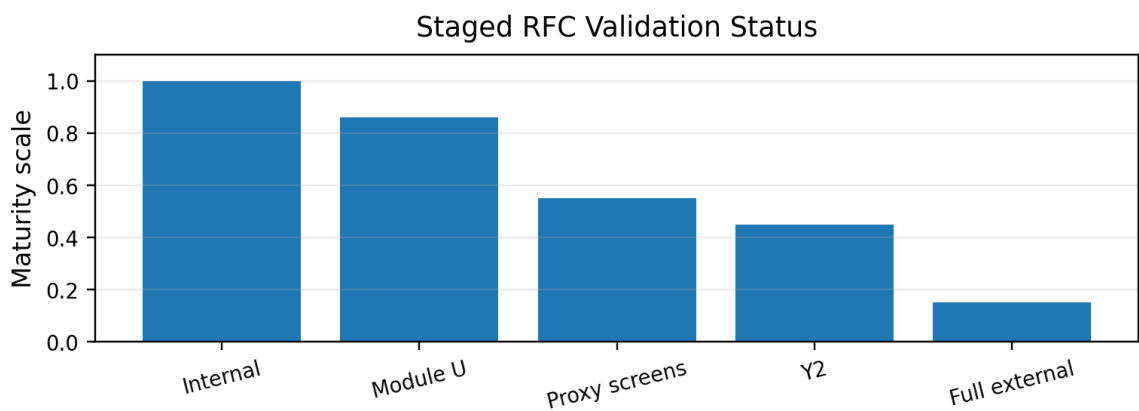


Figure 3. RFC validation is staged: internal derivation, Module U first-pass validation, partial/proxy screens, exploratory Y2, and future full external validation.

Module	Screen	Current status	Boundary
U	One-anchor constant table	completed first-pass	EM/atomic constants only
V	Precision cosmology	partial proxy	not BAO/SNe/CMB spectra
W	BBN light-abundance	partial proxy	Li7 unresolved; not full BBN network
X	CP/EDM bound	partial proxy	not full EFT/baryogenesis proof
Y2	Particle sector	exploratory candidate map	must be frozen/retested as Y3
Z	Observer/neural harness	model screen; data needed	not consciousness/EEG proof
QG	Spin-foam amplitude audit	finite audit passed	not full quantum-gravity proof

14.1 Module U: One-Anchor Constant Table Screen

Module U extends Modules S and T into a broader electromagnetic and atomic constant table without additional empirical anchors. It is the strongest new validation screen in this paper.

Metric	Value
independent constants scored	12
mean absolute percent error	0.07412%
median absolute percent error	0.07407%
minimum percent error	$9.64897 \times 10^{-6}\%$
maximum percent error	0.14827%
within 0.1%	8 / 12
within 1%	12 / 12
within 5%	12 / 12

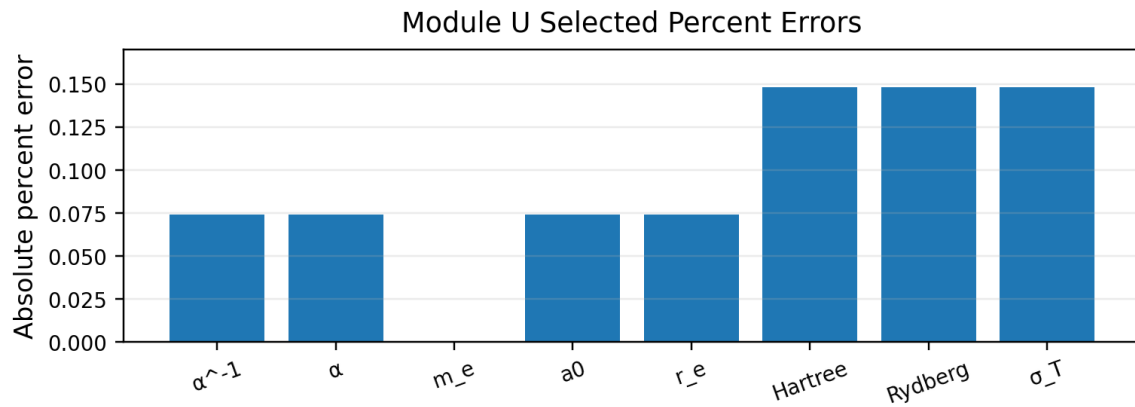


Figure 4. Selected Module U percent errors.

Constant	RFC	Reference	Error
α^{-1}	136.935	137.036	0.07405%
α	0.00730276	0.00729735	0.07410%
electron mass	9.10938 x 10 ⁻³¹ kg	9.10938 x 10 ⁻³¹ kg	9.64938 x 10 ⁻⁶ %
Bohr radius	5.28785 x 10 ⁻¹¹ m	5.29177 x 10 ⁻¹¹ m	0.07406%
Hartree energy	27.2517 eV	27.2114 eV	0.14827%
Rydberg energy	13.6259 eV	13.6057 eV	0.14827%
Thomson cross section	6.66232 x 10 ⁻²⁹ m ²	6.65246 x 10 ⁻²⁹ m ²	0.14824%
Bohr magneton	9.27401 x 10 ⁻²⁴ J/T	9.27401 x 10 ⁻²⁴ J/T	9.64897 x 10 ⁻⁶ %

Boundary: Module U is a first-pass one-anchor EM/atomic constant screen. It is not a derivation of every physical constant and does not cover G, hadron masses, weak-sector constants, or the full Standard Model hierarchy.

14.2 Module V: Precision Cosmology Compressed-Parameter Screen

Module V compares compressed RFC cosmology proxies against reference cosmological quantities. It is a useful first screen, but it is not a full precision-cosmology likelihood analysis.

Quantity	RFC/status	Reference	Error/status
H0 proxy	69.2176	67.4	2.69666%
Ω_m proxy	0.328608	0.315	4.31988%
Ω_Λ proxy	0.671392	0.685	1.98651%
σ_8 proxy	0.812251	0.811	0.15425%
n_s proxy	0.974317	0.965	0.96548%
r proxy	0.0108071	< 0.06	PASS
BAO	not full solver	dataset needed	not scored
SNe	not full solver	dataset needed	not scored
CMB spectra	not Boltzmann solver	CLASS/CAMB needed	not scored

Boundary: Module V is a compressed-parameter proxy. Full BAO likelihoods, SNe distance-modulus fitting, CMB spectra, Boltzmann-code comparison, and matter power spectrum validation remain future work.

14.3 Module W: BBN Light-Abundance Proxy Screen

Module W upgrades the earlier Yp-only proxy into a light-abundance comparison screen. The current proxy performs reasonably for Yp, D/H, and He3/H, while Li7 remains unresolved.

Abundance	RFC	Reference	Error
Yp	0.249278	0.245	1.74612%
D/H	2.45181×10^{-5}	2.55×10^{-5}	3.85043%

Abundance	RFC	Reference	Error
He3/H	1.01081×10^{-5}	1.0×10^{-5}	1.08071%
Li7/H	4.5×10^{-10}	1.6×10^{-10}	181.25%

Boundary: Module W is not a professional full BBN nuclear-reaction network. The lithium gap remains unresolved.

14.4 Module X: CP/EDM Bound Screen

Module X maps the frozen packet into CP/EDM bound proxies and baryon-eta comparison. It gives strong bound-screen behavior, while full EFT-level CP/EDM and baryogenesis calculations remain future work.

Item	RFC	Reference/bound	Status
electron EDM proxy	3.94987×10^{-39}	$< 1.1 \times 10^{-29} \text{ e cm}$	PASS
neutron EDM proxy	3.41089×10^{-31}	$< 1.8 \times 10^{-26} \text{ e cm}$	PASS
baryon eta proxy	6.0991×10^{-10}	6.1×10^{-10}	0.01481% error
legacy CKM phase proxy	-	-	157.12833% error before Y2
legacy PMNS phase proxy	-	-	82.00318% error before Y2

Boundary: Module X passes EDM bound proxies and gives a strong baryon eta proxy. It is not a full EFT-level CP/EDM, baryogenesis, CKM, or PMNS derivation.

14.5 Module Y2: Particle-Sector Refinement Candidate Map

Module Y2 was created to improve the high-error particle-sector outputs from the first-pass hierarchy screen. It introduces QCD/log-space dressing for quark masses, Wolfenstein-like structure for CKM, large-angle neutrino mixing architecture for PMNS, and modular phase comparison for CP phases.

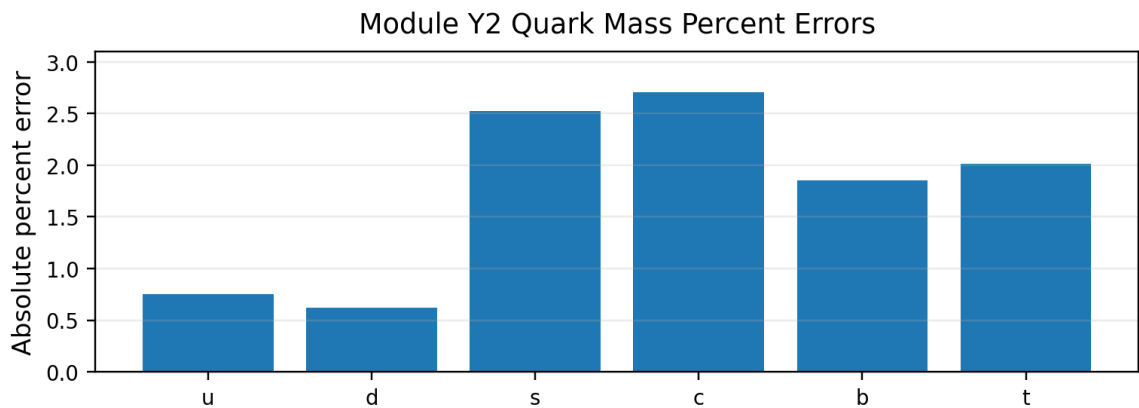


Figure 5. Module Y2 reduces quark mass errors to a mean of 1.74577% and a maximum of 2.70668%.

Quark	Y2 RFC MeV	Reference MeV	Error
u	2.17623	2.16	0.75117%
d	4.64097	4.67	0.62169%
s	95.3491	93	2.52591%
c	1235.63	1270	2.70668%
b	4102.52	4180	1.85351%
t	176242	172760	2.01567%

Sector	Best results	Remaining issue
Quark masses	mean error 1.74577%; max error 2.70668%	Exploratory map selected using references.
CKM	theta12 error 0.14407%; theta23 error 2.57199%; phase circular error 0.00981 degrees; sin-phase error 0.00674%	theta13 remains high at 28.23284%.

Sector	Best results	Remaining issue
PMNS	theta12 error 0.03822%; theta23 error 0.12940%; theta13 error 0.18808%; phase circular error 0.00187 degrees; mean angle error 0.11857%	Must freeze and retest as Y3.

Boundary: Module Y2 is exploratory candidate discovery, not final validation. Because Y2 uses reference values for map selection, the selected formulas must be frozen into Y3 and retested before independent validation claims are made.

14.6 Module Z: Observer, Branching, Neural, and EEG Validation Harness

Module Z separates internal observer/branching checks from external neural/EEG validation. The internal observer/branching checks pass; neural and EEG outputs are target signatures only until tested on real datasets.

Observer/branching item	RFC	Target	Status
observer divergence	7.49536×10^{-6}	$< 1e-5$	PASS
branch final/max	0.0599657	< 0.1	PASS
decoherence proxy	5.73826×10^{-8}	$< 1e-6$	PASS
memory correlation	0.873361	> 0.8	PASS

Neural/EEG target	RFC	Target	Error/status
neural fractal D	2.48944	2.45	1.60988%; needs EEG data
avalanche exponent	1.48944	1.5	0.70387%; needs neural data
spectral slope	1.97432	2.0	1.28415%; needs EEG data

Boundary: Module Z does not experimentally prove consciousness, observer branching, or many-worlds physics. It provides model-screen behavior and target signatures.

14.7 Module QG: Finite Spin-Foam Transition-Amplitude Audit

Module QG audits finite spin-foam transition-amplitude behavior from the frozen RFC packet. It is a finite audit, not a full quantum-gravity proof.

Spin-foam item	RFC	Status
finite amplitude $n = 18$	1.04394	finite screen
finite amplitude $n = 40$	1.04394	finite screen
relative tail	4.81169×10^{-12}	PASS

Spin-foam item	RFC	Status
unitarity proxy	1	PASS
refinement proxy	1	PASS
geometry coherence	0.940535	PASS

Boundary: Module QG is not a full transition-amplitude derivation, continuum theorem, gauge-invariance proof, renormalization result, or complete spin-foam formalism.

15. Reintegrating Prior RFC Modules as Downstream Projections

The revised framework does not discard the earlier module family. It reorganizes that family under a stricter dependency structure. Prior modules remain part of RFC, but they are not allowed to tune the upstream packet. They are downstream projections of the same triadic kernel.

This continuity matters because the earlier presentations are not obsolete in content; they are superseded in derivation architecture. The current framework reorganizes those materials under deterministic triadic closure and staged validation.

Older module family	Current interpretation
Cosmology and dark sector	Downstream QV compression/tail projection.
CP, baryogenesis, EDM	Downstream QV-CIF phase-bias projection; Module X bound/proxy screen.
BBN / abundance proxy	Module W light-abundance proxy; Li7 unresolved.
Mass and particle structure	Symbolic mass-energy projection plus exploratory Y2 particle-sector map.
Observer / decoherence / many-worlds	Downstream self-reference projection; Module Z model screen only.
Spin foam / quantum geometry	Finite QG audit and geometry coherence proxy; not proof.
Neural/fractal/cognition	Target signatures; real datasets needed.
Collapse-rebirth / global memory	Lagged recursive memory-transfer test.

The current public repository for this RFC rebuild is available here:

<https://github.com/Charlie-glitch83/Recursive-Fractal-Cosmology/tree/main>

16. What Is Proven Internally

The strongest current claim is not that RFC has proven every physical observable. The strongest claim is that RFC has become internally coherent as a deterministic triadic derivation system with a documented first-pass

downstream validation scaffold.

Category	Current result
Triad-first architecture	CIF, QV, and RFL function as the foundation, not as decorative framing.
Module G	The active packet is generated by deterministic triadic closure without MCMC/NUTS fitting.
Module R	Source-coupled global closure is QV-dominant and improves residual structure.
Module N V2	Dimensional identities are preserved: $\alpha_G \approx m^2$, $\alpha_{EME} \approx 1$, $\Lambda ET \approx 1$.
Downstream projection	Dark-sector, CP, observer, rebirth, and geometry screens consume the packet without retuning.
Dimensionless validation	All seven internal flags pass; overall validation score = 1.
Module S	One-anchor SI bridge is established.
Module T	Fine-structure-like dimensionless coupling map yields $\alpha_T^{-1} = 136.935$.
Module U	First-pass one-anchor EM/atomic constant table screen is completed.
Modules V/W/X/Z/QG	Partial/proxy/audit validation screens are documented.
Module Y2	Exploratory particle-sector map substantially improves quarks, CKM phase, and PMNS angles/phases, but must freeze/retest as Y3.

17. What Remains To Be Externally Validated

The staged validation screen should not be confused with full external validation. This paper explicitly separates completed first-pass screens, partial proxy screens, exploratory candidate maps, and future domain-complete validation.

Domain	Current status	Still needed
Physical constants	Module U completes a first-pass EM/atomic constant screen.	G, hadronic constants, weak-sector constants, full hierarchy, and independent reproduction.
Precision cosmology	Module V gives compressed H0, Ω_m , Ω_Λ , σ_8 , n_s , and r proxies.	BAO, SNe, CMB spectra, Boltzmann-code comparison, matter power spectrum.
BBN	Module W gives Y_p , D/H, He3/H, and Li7/H proxy.	Full nuclear reaction network; Li7 unresolved.
CP/EDM and baryogenesis	Module X passes EDM bound proxies and baryon eta proxy.	Full EFT mapping, EDM derivation, baryogenesis calculation.
Particle masses and mixing	Module Y2 improves quarks, CKM phase, and PMNS.	Freeze as Y3; solve CKM theta13; neutrino masses/splittings.
Observer / consciousness	Module Z passes internal model checks.	External measurement-model validation; no claim of consciousness proof.
Neural / EEG	Module Z gives target signatures.	Real EEG and neural avalanche dataset comparison without tuning.
Spin foam / QG	Module QG finite audit passes.	Full transition-amplitude derivation, continuum/refinement proof, gauge-invariance proof.
Reproduction	Repository scaffold assembled.	Independent external reproduction, code audit, peer review, archived release.

18. Discussion

The main achievement of the present paper is structural clarity. RFC no longer appears as a fitted bundle of modules. It is organized as a triad-first derivation system with a frozen packet and a staged validation scaffold.

The validation program is layered rather than binary. Module U is the strongest first-pass result because it extends the one-anchor SI bridge and Module T map into a 12-constant EM/atomic table. Modules V, W, X, Z, and QG are weaker but useful proxy and audit screens. Module Y2 is promising but must remain exploratory until frozen and retested.

This staged framing gives readers a clear map of what is internally established, what has been partially screened, what remains exploratory, and what still requires domain-complete external validation. That clarity is more credible than claiming final proof too early.

19. Conclusion

Recursive Fractal Cosmology: A Generative Ontology of Existence reconstructs RFC as a triad-based derivation framework rather than as a fitted modular model. Its central claim is that existence becomes physical through recursive interaction of CIF, QV, and RFL.

The active framework removes upstream fitting from the derivation chain and replaces it with deterministic triadic closure. Module R audits global closure. Module N V2 preserves dimensional identities. The downstream physical-projection and dimensionless validation layers test internal coherence. Module S introduces a one-anchor SI bridge. Module T maps an internal inverse-energy quantity into a dimensionless fine-structure-like value.

The validation-screen sequence U, V, W, X, Y2, Z, and QG extends the framework beyond internal reconstruction. Module U completes a first-pass one-anchor electromagnetic/atomic constant table. Modules V, W, X, Z, and QG provide partial compressed, proxy, or finite-audit screens. Module Y2 substantially improves the particle-sector map, but remains exploratory until frozen and retested as Y3.

RFC has therefore advanced from internal reconstruction to staged downstream validation. Full external validation, independent reproduction, peer review, and domain-complete datasets remain future work.

20. Appendices

The repository appendix has been expanded into a validation and reproducibility protocol. The revised appendix sequence is summarized below.

Appendix	Content
A	Current Canonical RFC Architecture
B	Module G Deterministic Triadic Closure
C	Deprecated Historical Module G
D	Shared Recursive Kernel
E	Module R Audit
F	Module N V2 Dimensional Projection
G	Downstream Physical-Projection Screen
H	Dimensionless Validation Layer
I	Module S One-Anchor SI Bridge
J	Module T Coupling Map
K	Module U Constant Table
L	Module V Cosmology Screen
M	Module W BBN Screen
N	Module X CP/EDM Screen
O	Module Y2 Particle-Sector Refinement
P	Module Z Observer/Neural Harness
Q	Module QG Spin-Foam Audit
R	Reintegrated Prior Modules
S	Glossary of Symbolic Fields
T	JSON Configuration and Reproducibility Protocol
U	Claim Boundary Table
V	Revision Summary

Appendix	Content
W	Final Boundary Statement

21. Bibliography and Reference Strategy

This paper does not present the final external bibliography as a proof of validation. It defines the reference categories that future full validation must use. References contextualize comparison; they do not originate the frozen RFC packet.

Reference class	Use in RFC validation
CODATA / NIST constants	Comparison layer for Module U and future physical-constant tests.
Particle Data Group	Reference layer for particle masses, CKM, PMNS, and sector-specific particle screens.
Planck / BAO / SNe / CMB literature	Future full precision-cosmology comparison beyond Module V.
BBN abundance and nuclear-rate literature	Future professional network comparison beyond Module W.
EDM and baryogenesis literature	Future EFT-level validation beyond Module X.
Neural avalanche and EEG fractal literature	External data layer for Module Z target signatures.
Spin-foam and loop quantum gravity literature	Formal comparison layer for Module QG.
Reproducibility and code-audit standards	External validation and independent reproduction.

In future versions, each comparison class should be converted into a conventional reference list with explicit equations, datasets, uncertainty conventions, and reproducibility instructions. The key rule remains unchanged: the frozen RFC packet is generated first; external references are used afterward for comparison, except where a module is explicitly labeled exploratory, such as Y2.

22. Final Boundary Statement

Status layer	Current statement
Internal derivation and coherence	established
First-pass downstream validation scaffold	partially established
Exploratory particle-sector refinement	advanced through Y2 but not independently validated
Full external empirical validation	future work

The current RFC status can therefore be stated without overclaiming: RFC is internally coherent as a deterministic triadic derivation system with a growing first-pass validation scaffold. It is not yet fully externally validated as a complete laboratory theory of all constants and observables.